Variational Autoencoders: From VAE to Conditional VAE

A Visual and Mathematical Introduction

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March 2026

Abstract

This note provides a self-contained introduction to the Variational Autoencoder (VAE) and its conditional extension (CVAE). We cover the probabilistic formulation, the evidence lower bound (ELBO), the reparameterization trick, and training procedures for both models, with illustrative diagrams throughout.

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1 Background: Latent Variable Models

Many real-world datasets (images, text, audio) lie on low-dimensional manifolds embedded in high-dimensional observation spaces. *Latent variable models* capture this by introducing unobserved variables z that explain the data x:

$$p_\theta(x) = \int p_\theta(x|z)\, p(z)\, dz.$$

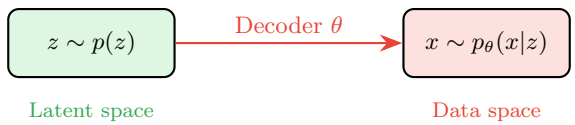

Figure 1: The generative process: sample z from prior, then generate x via the decoder.

The generative story:

1. Sample a latent code $z \sim p(z)$ (the *prior*, typically $\mathcal{N}(0, I)$).
2. Generate data $x \sim p_\theta(x|z)$ (the *decoder / likelihood*).

Two fundamental challenges arise:

- **Intractable marginal likelihood.** $p_\theta(x) = \int p_\theta(x|z)p(z)\, dz$ is intractable when $p_\theta(x|z)$ is a neural network.
- **Intractable posterior.** $p_\theta(z|x) = p_\theta(x|z)p(z)/p_\theta(x)$ requires $p_\theta(x)$.

The **VAE** addresses both via *amortized variational inference*.

2 Variational Autoencoder (VAE)

2.1 Overall Architecture

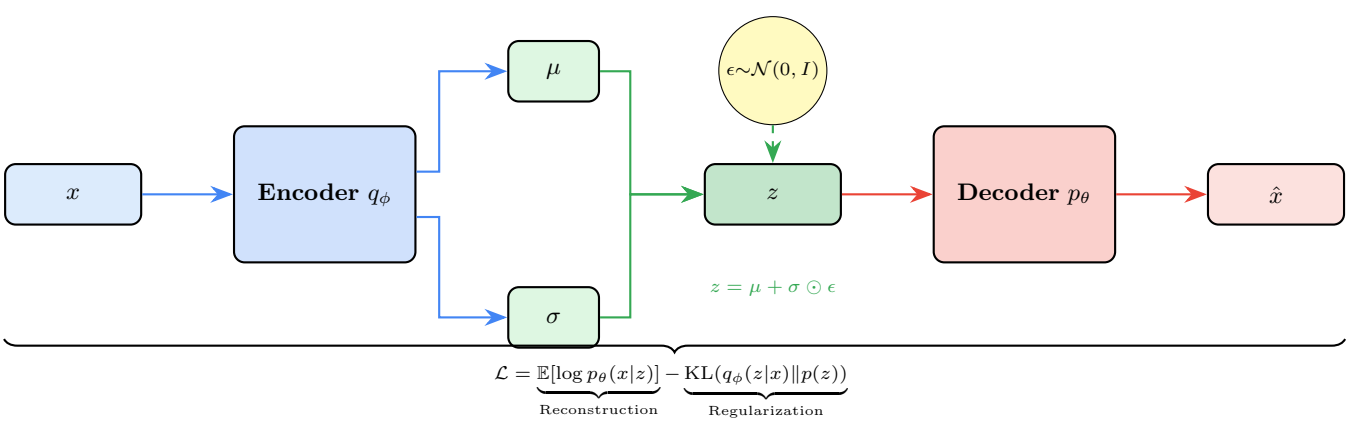

Figure 2: VAE architecture. The encoder maps x to distribution parameters (μ, σ). A latent code z is sampled via the reparameterization trick. The decoder reconstructs $\hat{x}$ from z. Training maximizes the ELBO.

2.2 Model Specification

Component	Distribution	Role
Prior	$p(z) = \mathcal{N}(z; 0, I)$	Latent space structure
Decoder	$p_\theta(x\|z)$	Maps latent codes to data
Encoder	$q_\phi(z\|x) = \mathcal{N}(z; \mu_\phi(x), \mathrm{diag}(\sigma^2_\phi(x)))$	Approximates posterior

2.3 Evidence Lower Bound (ELBO)

Since $\log p_\theta(x)$ is intractable, we derive a tractable lower bound.

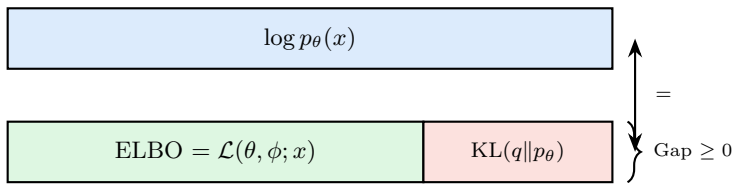

Figure 3: Decomposition: $\log p_\theta(x) = \mathrm{ELBO} + \mathrm{KL}(q_\phi(z|x)\|p_\theta(z|x))$. Since $\mathrm{KL} \geq 0$, the ELBO is a lower bound.

$$
\begin{aligned}
\log p_\theta(x) &= \log \int p_\theta(x, z)\, dz \\
&= \log \int q_\phi(z|x) \frac{p_\theta(x, z)}{q_\phi(z|x)}\, dz \\
&\geq \int q_\phi(z|x) \log \frac{p_\theta(x, z)}{q_\phi(z|x)}\, dz \qquad \text{(Jensen's inequality)} \\
&= \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \mathrm{KL}(q_\phi(z|x) \,\|\, p(z))\,. \qquad (1)
\end{aligned}
$$

$$
\boxed{\mathcal{L}(\theta, \phi; x) = \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]}_{\text{Reconstruction}} - \underbrace{\mathrm{KL}(q_\phi(z|x)\|p(z))}_{\text{Regularization}}}
$$

Precisely, we have:

$$
\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + \mathrm{KL}(q_\phi(z|x)\|p_\theta(z|x)) \geq \mathcal{L}(\theta, \phi; x).
$$

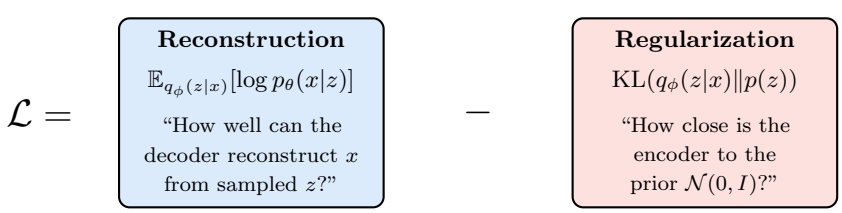

Figure 4: The two terms of the ELBO and their intuitive meanings.

Interpretation of each term (Figure 4).

2.4 The Reparameterization Trick

To optimize the ELBO via gradient descent, we need gradients w.r.t. ϕ. However, $\mathbb{E}_{q_\phi(z|x)}[\cdot]$ involves sampling from a ϕ-dependent distribution.

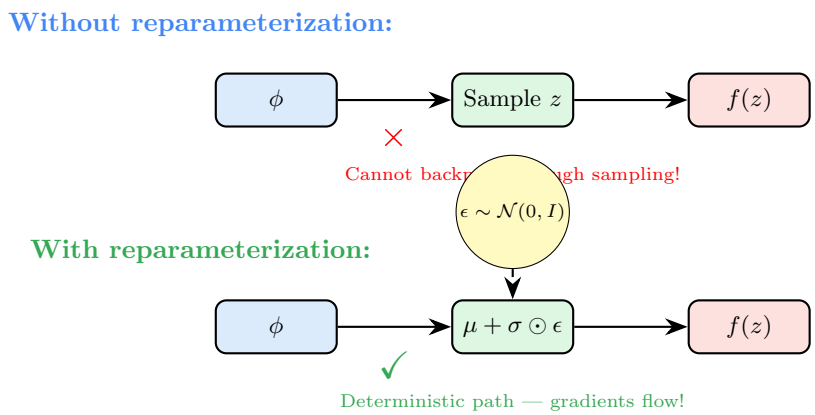

Figure 5: The reparameterization trick moves stochasticity to an external noise variable ϵ, creating a differentiable path from ϕ to z.

The trick expresses z as a deterministic, differentiable function of ϕ:

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

Now gradients w.r.t. ϕ flow through μ_ϕ and σ_ϕ via standard backpropagation.

2.5 Closed-Form KL for Gaussians

When $q_\phi(z|x) = \mathcal{N}(\mu, \mathrm{diag}(\sigma^2))$ and $p(z) = \mathcal{N}(0, I)$ with $z \in \mathbb{R}^d$:

$$\mathrm{KL}(q_\phi(z|x) \| p(z)) = \frac{1}{2} \sum_{j=1}^{d} \left(\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2 \right).$$

2.6 Graphical Model

(a) Generative model p_θ

(b) Inference model q_ϕ

Figure 6: Graphical models for VAE. Shaded nodes are observed; unshaded are latent. (a) The generative model defines $p(z)p_\theta(x|z)$. (b) The inference model (encoder) approximates $p_\theta(z|x)$.

2.7 Training Algorithm

Algorithm 1: VAE Training

Input: Dataset $\mathcal{D} = \{x^{(1)}, \ldots, x^{(N)}\}$, learning rate η

Repeat until convergence:

1. Sample mini-batch $\{x^{(i)}\}_{i=1}^{M}$ from $\mathcal{D}$
2. **For** each $x^{(i)}$ in mini-batch:
   (a) **Encode**: $\mu^{(i)}, \sigma^{(i)} \leftarrow \text{Encoder}_\phi(x^{(i)})$
   (b) **Sample**: $\epsilon^{(i)} \sim \mathcal{N}(0, I)$; $z^{(i)} = \mu^{(i)} + \sigma^{(i)} \odot \epsilon^{(i)}$
   (c) **Decode**: Compute $\log p_\theta(x^{(i)}|z^{(i)})$
   (d) **KL**: $D_i = \frac{1}{2} \sum_j \left((\mu_j^{(i)})^2 + (\sigma_j^{(i)})^2 - 1 - \log(\sigma_j^{(i)})^2 \right)$
3. $\mathcal{L} \leftarrow \frac{1}{M} \sum_{i=1}^{M} \left[\log p_\theta(x^{(i)}|z^{(i)}) - D_i \right]$
4. $\theta, \phi \leftarrow \theta, \phi + \eta \nabla_{\theta,\phi} \mathcal{L}$ (e.g., Adam)

2.8 Generation (Sampling)

After training:

1. Sample $z \sim \mathcal{N}(0, I)$.
2. Compute $\hat{x} = \mu_\theta(z)$ (or sample $\hat{x} \sim p_\theta(x|z)$).

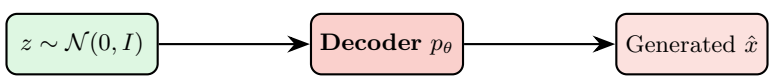

Figure 7: Generation with a trained VAE: sample from prior, then decode.

3 Conditional VAE (CVAE)

3.1 Motivation

The standard VAE models $p(x)$. In many applications, we want $p(x|c)$ where c is a *condition*:

- Generate handwritten digits conditioned on class $c \in \{0, \ldots, 9\}$.
- Generate face images conditioned on attributes.
- Predict multiple plausible futures conditioned on context.

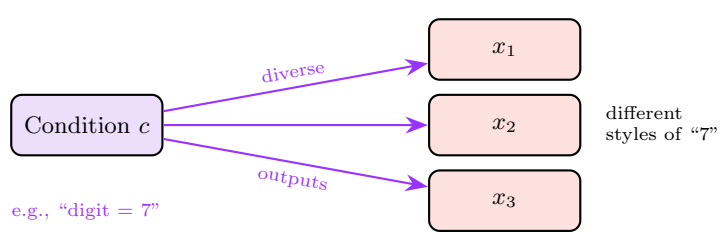

Figure 8: CVAE generates diverse outputs x given a condition c. The latent variable z captures the variation not explained by c.

3.2 Overall Architecture

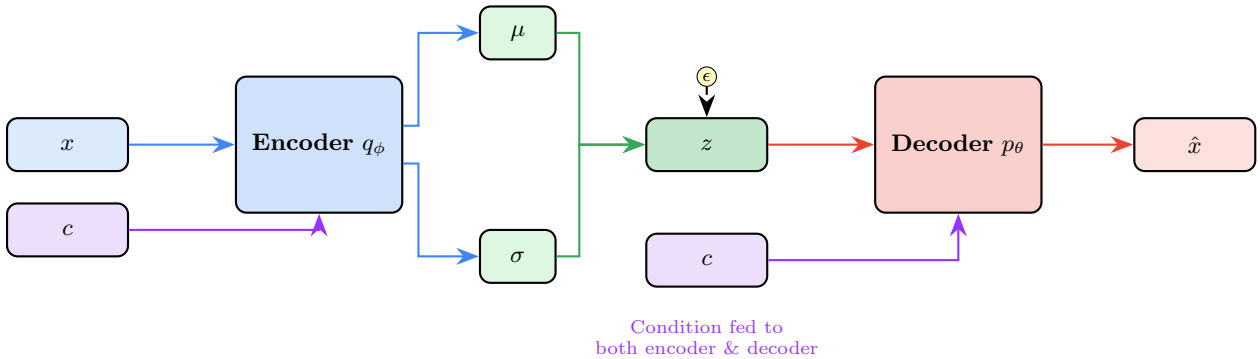

Figure 9: CVAE architecture. Both encoder and decoder receive the condition c as additional input. The encoder is $q_\phi(z|x,c)$ and the decoder is $p_\theta(x|z,c)$.

3.3 Model Specification

Component	Distribution	Role
Conditional prior	$p_\alpha(z\|c)$ or simply $\mathcal{N}(0,I)$	Latent structure given condition
Conditional decoder	$p_\theta(x\|z,c)$	Generates data from z + condition
Conditional encoder	$q_\phi(z\|x,c)$	Infers latent code from data + condition

Remark 1. *A common simplification is* $p(z|c) = p(z) = \mathcal{N}(0,I)$. *A* learned *conditional prior* $p_\alpha(z|c) = \mathcal{N}(\mu_\alpha(c), \mathrm{diag}(\sigma_\alpha^2(c)))$ *is more expressive and allows different conditions to occupy different latent regions.*

3.4 ELBO Derivation for CVAE

All distributions are conditioned on c:

$$\begin{aligned}\log p_\theta(x|c) &= \log \int p_\theta(x|z,c)\, p_\alpha(z|c)\, dz \\ &\geq \mathbb{E}_{q_\phi(z|x,c)}[\log p_\theta(x|z,c)] - \mathrm{KL}(q_\phi(z|x,c) \,\|\, p_\alpha(z|c))\,. \end{aligned} \tag{2}$$

$$\boxed{\mathcal{L}_{\mathrm{CVAE}}(\theta,\phi,\alpha;x,c) = \underbrace{\mathbb{E}_{q_\phi(z|x,c)}[\log p_\theta(x|z,c)]}_{\text{Conditional reconstruction}} - \underbrace{\mathrm{KL}(q_\phi(z|x,c)\|p_\alpha(z|c))}_{\text{Conditional regularization}}}$$

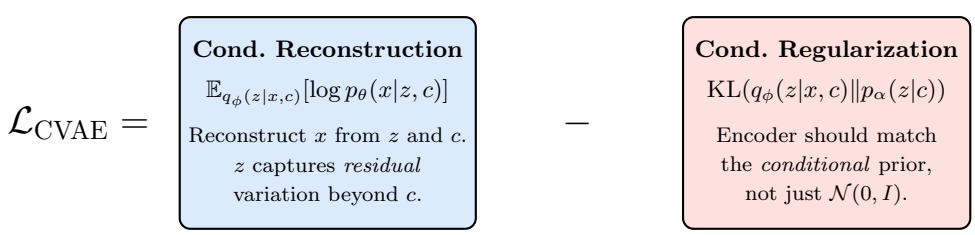

Figure 10: The two terms of the CVAE ELBO.

3.5 Graphical Model

(a) $p_\alpha(z|c)\, p_\theta(x|z,c)$ (b) $q_\phi(z|x,c)$

Figure 11: Graphical models for CVAE. Purple nodes are conditions (always observed); gray nodes are observed data; white nodes are latent.

3.6 Training Algorithm

Algorithm 2: CVAE Training

Input: Dataset $\{(x^{(i)}, c^{(i)})\}_{i=1}^{N}$, learning rate η

Repeat until convergence:

1. Sample mini-batch $\{(x^{(i)}, c^{(i)})\}_{i=1}^{M}$
2. **For** each $(x^{(i)}, c^{(i)})$ in mini-batch:
   (a) **Encode**: $\mu^{(i)}, \sigma^{(i)} \leftarrow \text{Encoder}_\phi(x^{(i)}, c^{(i)})$
   (b) **Sample**: $z^{(i)} = \mu^{(i)} + \sigma^{(i)} \odot \epsilon^{(i)},\ \epsilon^{(i)} \sim \mathcal{N}(0, I)$
   (c) **Decode**: Compute $\log p_\theta(x^{(i)}|z^{(i)}, c^{(i)})$
   (d) **KL**: Compute $\text{KL}(q_\phi(z|x^{(i)}, c^{(i)}) \| p_\alpha(z|c^{(i)}))$
3. $\mathcal{L} \leftarrow \frac{1}{M} \sum_i \left[\log p_\theta(x^{(i)}|z^{(i)}, c^{(i)}) - \text{KL}(\cdots)\right]$
4. Update $\theta, \phi, \alpha \leftarrow \theta, \phi, \alpha + \eta \nabla \mathcal{L}$

3.7 Conditional Generation

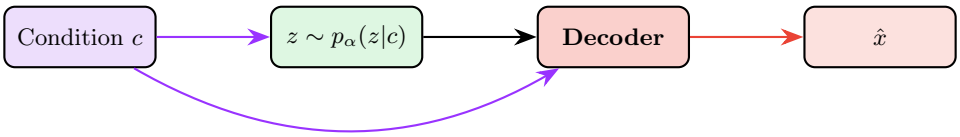

Figure 12: Conditional generation: given c, sample z from conditional prior, then decode with both z and c.

4 VAE vs. CVAE: Side-by-Side Comparison

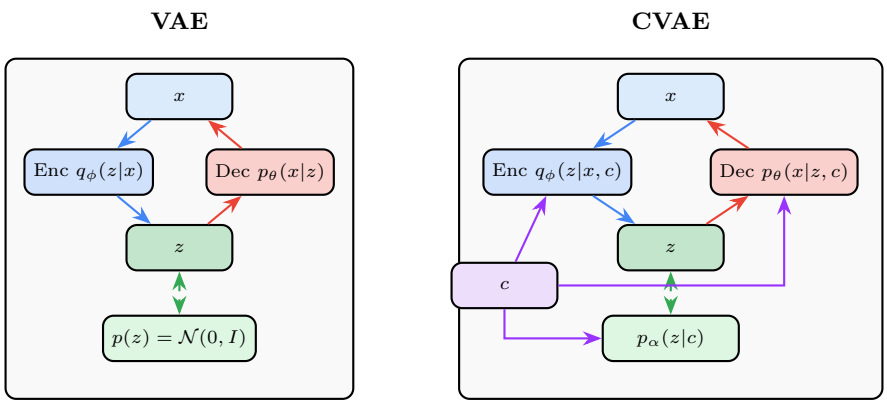

Figure 13: Side-by-side comparison of VAE (left) and CVAE (right). The key difference: CVAE feeds condition c to encoder, decoder, and optionally the prior.

	VAE	**CVAE**
Goal	Model $p(x)$	Model $p(x\|c)$
Prior	$p(z) = \mathcal{N}(0, I)$	$p_\alpha(z\|c)$ (or $\mathcal{N}(0, I)$)
Encoder input	x	(x, c)
Decoder input	z	(z, c)
Latent z captures	All variation in x	Residual variation beyond c
ELBO	$\mathbb{E}_q[\log p(x\|z)] - \mathrm{KL}(q\|p)$	$\mathbb{E}_q[\log p(x\|z, c)] - \mathrm{KL}(q\|p_c)$

4.1 When to Use Which?

- **VAE**: No conditioning information available; want to learn the full data distribution $p(x)$.
- **CVAE**: Have auxiliary information c and want controlled/conditional generation:
  - Generate specific digit classes.
  - Predict diverse futures given context.
  - Conditional data augmentation.

5 Common Challenges and Solutions

5.1 Posterior Collapse

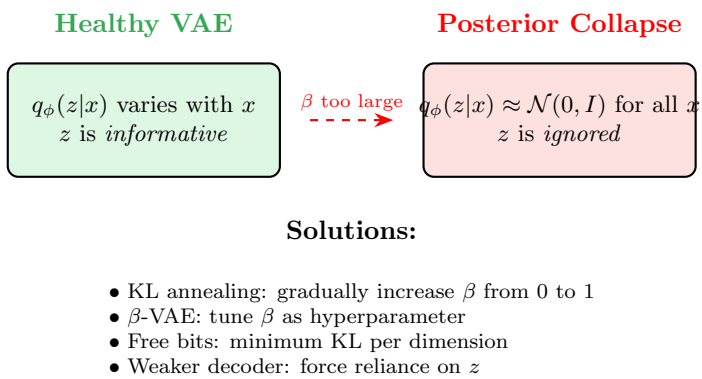

Figure 14: Posterior collapse: the encoder ignores input and outputs the prior for all x, making z uninformative. Several remedies exist.

5.2 Blurry Samples

VAE samples tend to be blurrier than GAN samples because:

- Gaussian likelihood with MSE loss averages over modes.
- The approximate posterior may not capture the true posterior well.

Solutions: More expressive decoders, normalizing flows in latent space, or hybrid VAE-GAN models.

5.3 The β-VAE Trade-off

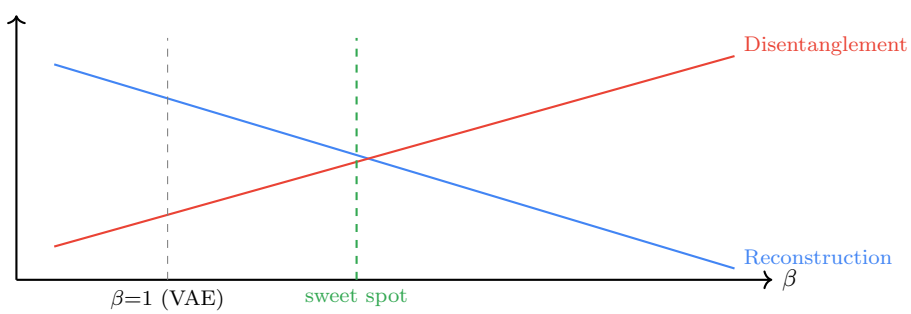

Figure 15: The β-VAE trade-off: increasing β improves disentanglement but degrades reconstruction quality.

6 Summary

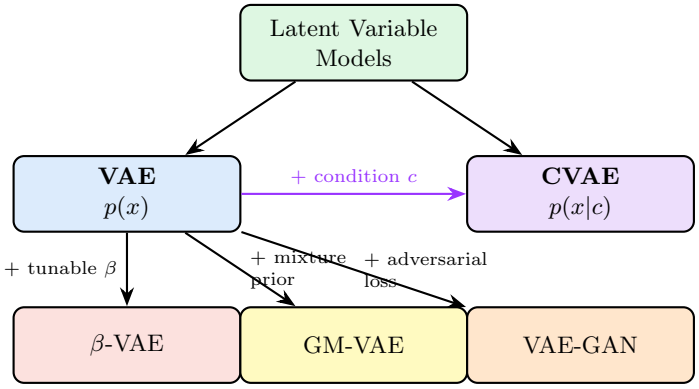

Figure 16: The VAE family: VAE and CVAE are the two core frameworks, with many extensions.

- **VAE** provides a principled framework for learning deep latent variable models by jointly training an encoder and decoder via the ELBO.
- The **reparameterization trick** enables end-to-end gradient-based optimization through stochastic sampling.
- **CVAE** extends VAE to conditional generation by incorporating condition c into all components.
- Both models optimize a variational lower bound on the (conditional) log-likelihood.

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